

3 Thermodynamic potentials – solutions

The problems are roughly in order of difficulty. Those indicated with ♡ are those which it is most important to work on. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

Problem 3.1 Rubber: a more complicated, and interesting, internal energy

- (i) The energy differential is the sum of TdS and the two types of work. These are usually obtained by multiplying the intensive parameter with the differential of the associated extensive parameter. This gives $dU = TdS - pdV + fdL$
- (ii) The energy U is a function of the extensive parameters, here S , V and L . The thermodynamic potentials that correspond to a choice of variables are (their names, H' , H and G are somewhat arbitrary)

$$\begin{aligned} H' &= U - fL , \\ H &= U + pV - fL , \\ G &= U - TS + pV - fL . \end{aligned}$$

Their differentials are

$$\begin{aligned} dH' &= TdS - pV - Ldf , \\ dH &= TdS + Vdp - Ldf , \\ dG &= -SdT + Vdp - Ldf . \end{aligned}$$

Let us construct the first one in more detail. The internal energy, U , is a function of (S, V, L) , and we want (S, V, f) . So one must replace L with f . This is achieved by introducing a new thermodynamic potential H' (the name has no convention) with the Legendre transformation $H' = U - fL$. Note that H' involves only functions of state (U , f and L), so it is a function of state itself.

If we differentiate H' we find

$$\begin{aligned} dH' &= dU - d(fL) \\ &= TdS - pdV + fdL - f dL - Ldf \\ &= TdS - PdV - Ldf , \end{aligned}$$

so H' is indeed a function of (S, V, f) . The differential only contains functions of state, so it will hold for any type of transformation, whether reversible or irreversible.

- (iii) To derive the Maxwell relations, one must determine the variables of interest. These are both the variable of differentiation, and the variables that are explicitly kept constant. For example, for the partial differential

$$\left(\frac{\partial T}{\partial V} \right)_{L,S} ,$$

it is V , L and S . Therefore for the three Maxwell relations in the question, we need to derive the thermodynamic potentials that have natural variables (S, V, L) (that is actually U !), (T, p, L) and (S, p, L) . The last two are novel. They are constructed in a similar fashion to the example above. One finds

$$\begin{aligned}\left(\frac{\partial T}{\partial V}\right)_{L,S} &= -\left(\frac{\partial p}{\partial S}\right)_{V,L}, \\ \left(\frac{\partial S}{\partial L}\right)_{T,p} &= -\left(\frac{\partial f}{\partial T}\right)_{p,L}, \\ \left(\frac{\partial S}{\partial f}\right)_{p,L} &= \left(\frac{\partial L}{\partial T}\right)_{S,p}.\end{aligned}$$

In the last relation, note that one would want to use the variables p , f and L . This is impossible, because both f and L appear in the same term in the differential of the energy. Instead, start with

$$\left(\frac{\partial f}{\partial S}\right)_{p,L}, \text{ then use } \left(\frac{\partial x}{\partial y}\right)_z = \left[\left(\frac{\partial y}{\partial x}\right)_z\right]^{-1}.$$

- (iv) ♣ Poisson's Ratio of rubber is almost exactly $1/2$ (see my Latex notes on Temperature and Matter section 4.1.1), which implies it does not change volume when stretched. (I.e. the lateral contraction almost exactly compensates for the extension to reserve the volume.) Thus regarding L and V as independent variables is especially accurate.

Problem 3.2 Gibbs-Helmholtz

Starting on RHS:

$$\begin{aligned}-T^2 \left(\frac{\partial}{\partial T}\right)_V \left(\frac{F}{T}\right) &= -T^2 \left(\frac{1}{T} \left(\frac{\partial F}{\partial T}\right)_V - \frac{F}{T^2}\right) \\ &= -T \left(\frac{\partial F}{\partial T}\right)_V + F \\ &= TS + F \\ &= U.\end{aligned}\tag{3.1}$$

In simplifying Eq. (3.1) we use

$$-\left(\frac{\partial F}{\partial T}\right)_V = S.$$

Problem 3.3 Equilibration problem

$$dS = dS_1 + dS_2 = \frac{dU_1}{T_1} + \frac{p_1 dV_1}{T_1} + \frac{dU_2}{T_2} + \frac{p_2 dV_2}{T_2}.$$

Because $dV_1 = dV_2 = 0$ and $dU_1 = -dU_2$, the equilibrium condition $dS = 0$ implies $T_1 = T_2$.

Problem 3.4 The World at constant pressure

(i) Equating the coefficients of the differentials in

$$dG = \left(\frac{\partial G}{\partial T} \right)_p dT + \left(\frac{\partial G}{\partial p} \right)_T dp$$

and $dG = -SdT + Vdp$,

we see:

$$S = - \left(\frac{\partial G}{\partial T} \right)_p \quad \text{and} \quad V = \left(\frac{\partial G}{\partial p} \right)_T. \quad (3.2)$$

Then lets us cross differentiate Eq. (3.2) (using $\partial_{xy}^2 f = \partial_{yx}^2 f$):

$$- \left(\frac{\partial S}{\partial p} \right)_T = \left(\frac{\partial}{\partial p} \right)_T \left(\frac{\partial G}{\partial T} \right)_p = \left(\frac{\partial}{\partial T} \right)_p \left(\frac{\partial G}{\partial p} \right)_T = \left(\frac{\partial V}{\partial T} \right)_p. \quad (3.3)$$

(ii) Using

$$C_p = T \left(\frac{\partial S}{\partial T} \right)_p,$$

one finds

$$\begin{aligned} \left(\frac{\partial C_p}{\partial p} \right)_T &= T \left[\frac{\partial}{\partial p} \left(\frac{\partial S}{\partial T} \right)_p \right]_T \\ &= T \left[\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial p} \right)_T \right]_p \\ &= -T \left(\frac{\partial^2 V}{\partial T^2} \right)_p, \end{aligned}$$

using Eq. (3.3).

(iii) Apply the last expression to

$$V = \frac{nRT}{p} + nb$$

and find that

$$\left(\frac{\partial C_p}{\partial p} \right)_T = 0,$$

that is to say C_p is independent of p .

♣ Problem 3.5 Hard labour

The first thing to notice is that as expected the compressibility of liquid water is minuscule. Therefore even though the volume will change during compression, we can assume that it will stay very close to the volume at 1 atm (1 l/kg, so here $V_0 = 10^{-4} \text{ m}^3$).

(i) Using

$$\kappa \simeq -\frac{1}{V_0} \left(\frac{\partial V}{\partial p} \right)_T,$$

one gets that at constant T , $dV = -\kappa V_0 dp$, and integrating

$$\delta W = - \int p dV \Rightarrow W = \kappa V_0 \int_{p_1}^{p_2} p dp \simeq \frac{\kappa V_0 p_2^2}{2} \simeq 25 \text{ mJ} .$$

(ii) From question 4 part (i),

$$\left(\frac{\partial S}{\partial p} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_p ,$$

therefore at constant T , $dS = V\beta dp \simeq V_0\beta dp$ and $\delta Q = T dS = TV_0\beta dp$, which once integrated gives $Q \simeq TV_0\beta p_2 \simeq 44 \text{ J}$.

(iii) From the first law, $\Delta U = W - Q \simeq -Q$.

(iv) For the final part, we should consider temperature as a function of pressure and entropy, so

$$dT = \left(\frac{\partial T}{\partial p} \right)_S dp + \left(\frac{\partial T}{\partial S} \right)_p dS . \quad (3.4)$$

Thus since $dS = 0$ for an adiabatic process, we have

$$dT = \left(\frac{\partial T}{\partial p} \right)_S dp .$$

Using the cyclic derivatives identity and again the Maxwell relation above, one finds

$$\left(\frac{\partial T}{\partial p} \right)_S = - \left(\frac{\partial T}{\partial S} \right)_p \left(\frac{\partial S}{\partial p} \right)_T = \frac{T}{C_p} \left(\frac{\partial V}{\partial T} \right)_p \simeq \frac{T}{C_p} V_0 \beta . \quad (3.5)$$

As a result substituting Eq. (3.5) into Eq. (3.4) we see:

$$\frac{dT}{T} = \frac{V_0 \beta}{C_p} dp \quad \text{and} \quad T_2 = T_1 \exp \left(\frac{V_0 \beta}{C_p} p_2 \right) .$$

The heat capacity is $C_p = 4200 \text{ J/K}$, which gives an increase in temperature of 0.010 K. This is pretty much what you get if you assume that the heat calculated previously serves to heat up the water. (Why is that by the way?)

♣ Problem 3.6 Pressure squeezed from Clapeyron

The Clapeyron equation is

$$\frac{dp(T)}{dT} = \frac{nL}{T\Delta V} .$$

Since the volume of the gas is much larger than the volume of the liquid, ΔV is (almost) the volume V of the gas, and therefore

$$\frac{dp(T)}{dT} = \frac{Lp}{RT^2} ,$$

using the equation of state for the ideal gas.

This can be integrated to

$$p = p_0 \exp \left(-\frac{L}{RT} \right) ,$$

where p_0 is a constant.